

Summer Math Adventure

W5
D1

Week 5 · Day 1 | Angles: Types, Measuring & Drawing

Name: _____

Date: _____

GEOMETRY: ANGLES

Angle Types: Acute $< 90^\circ$ | Right $= 90^\circ$ | Obtuse: 90° – 180° | Straight $= 180^\circ$ | Reflex: 180° – 360°

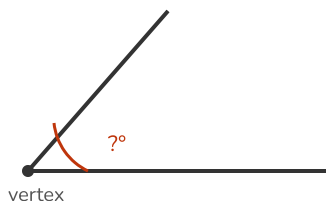
Using a protractor: Place center hole on vertex. Line up one ray on 0° . Read the scale where the other ray crosses.

Remember: Acute $\rightarrow$ use the SMALLER number; Obtuse $\rightarrow$ use the LARGER number on the protractor.

Part A – Name That Angle

Identify each angle type and estimate the measure. Then use your protractor to measure exactly.

Angle 1

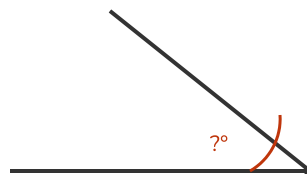


Type:

Measured:

°

Angle 2

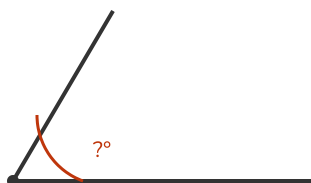


Type:

Measured:

°

Angle 3

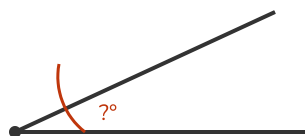


Type:

Measured:

°

Angle 4



Type:

Measured:

°

Part B – Draw the Angle

Use your protractor and the starting ray given. Draw the second ray to form the angle.

Draw a **35°** angle.

Draw a **110°** angle.

Draw a **75°** angle.

Angles

Summer Math Adventure

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D2

Week 5 · Day 2 | Lines, Triangles & Quadrilaterals

Name: _____

Date: _____

GEOMETRY: SHAPES

Line types: Line (infinite both ways), Ray (one endpoint), Line Segment (two endpoints)

Parallel lines never meet. Perpendicular lines meet at 90° .

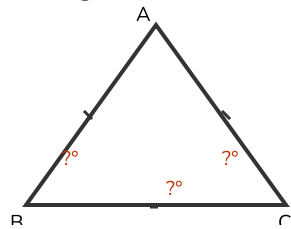
Triangle angles always add to 180° . Quadrilateral angles always add to 360° .

Triangle types: Equilateral (3 equal), Isosceles (2 equal), Scalene (none equal), Right (one 90° angle)

Part A – Classify the Triangle

Use your protractor to measure the angles. Then classify by angles AND by sides.

Triangle 1



Angle A:

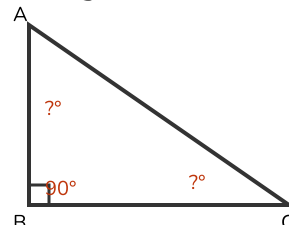
Angle B:

Angle C:

Type (sides):

Type (angles):

Triangle 2



Angle A:

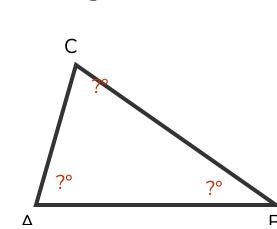
Angle B: 90°

Angle C:

Type (sides):

Type (angles):

Triangle 3



Angle A:

Angle B:

Angle C:

Type (sides):

Type (angles):

Part B – Find the Missing Angle

1. Triangle: angles are 48° , 75° , and $?$

$?$ =

o

2. Triangle: angles are 90° , 36° , and $?$

$?$ =

o

3. Quadrilateral: angles are 85° , 110° , 95° , and $?$

o

4. Parallelogram: one angle is 68° . Find all 4

angles

Name: _____

Date: _____

AREA & PERIMETER

Perimeter = total distance around the outside (add all sides). Units: cm, m, ft, in.

Area = space inside a shape. Units: cm², m², ft², in².

- Rectangle: $A = \text{length} \times \text{width}$ $P = 2(l + w)$
- Triangle: $A = \frac{1}{2} \times \text{base} \times \text{height}$
- Parallelogram: $A = \text{base} \times \text{height}$ (height is perpendicular distance, NOT the slanted side!)

Part A – Find Area & Perimeter

Shape 1 – Rectangle



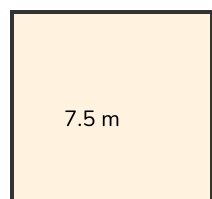
Perimeter =

cm

Area =

cm²

Shape 3 – Square



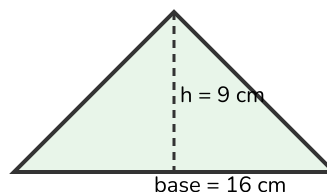
Perimeter =

m

Area =

m²

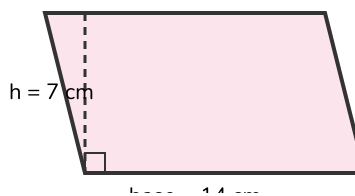
Shape 2 – Triangle



Area =

cm²

Shape 4 – Parallelogram



Area =

cm²

Part B – Composite Shapes

Summer Math Booklet: Grade 4 to 5, Week 5 Day 3
Break the shape into rectangles or triangles, find each area, then add.

Area & Perimeter

Find the area of the L-shaped figure:

10 ft

Summer Math Adventure

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D4

Week 5 · Day 4 | Volume of Rectangular Prisms & Coordinate Plane

Name: _____

Date: _____

VOLUME & COORDINATES

Volume = length $\times$ width $\times$ height. Units: cm^3 , m^3 , ft^3 , in^3 .

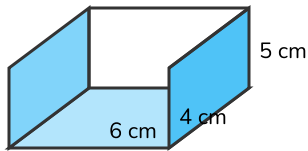
Think of it as: (base area) $\times$ height, or count cubic unit cubes.

Coordinate plane: $(x, y) \rightarrow x = \text{right/left}, y = \text{up/down}$. Origin = $(0,0)$.

Quadrant I: $(+, +)$ II: $(-, +)$ III: $(-, -)$ IV: $(+, -)$

Part A – Volume of Rectangular Prisms

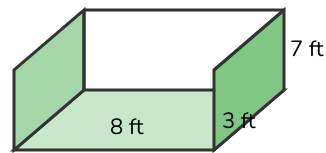
Prism 1



$$V = l \times w \times h = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} =$$

_____ cm^3

Prism 2



$$V =$$

_____ ft^3

Word Problem: A fish tank is 24 inches long, 12 inches wide, and 18 inches tall. What is its volume?

$$V =$$

_____ in^3

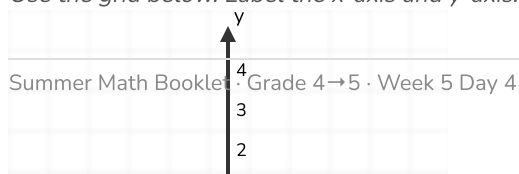
Find the missing dimension: A box has volume 360 cm^3 . Its base is $12 \text{ cm} \times 6 \text{ cm}$. What is the height?

$$\text{Height} =$$

_____ cm

Part B – Coordinate Plane

Use the grid below. Label the x-axis and y-axis. Plot each point and label it.



Plot and label these points:

A $(3, 2)$ B $(-1, 4)$ C $(-3, -2)$

D $(2, -3)$ E $(0, 3)$ F $(-4, 0)$

Summer Math Adventure

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D5

Week 5 · Day 5 | FULL CUMULATIVE REVIEW – All 5 Weeks

Name: _____

Date: _____

CUMULATIVE REVIEW

You've covered: Place value · Rounding · Add/Subtract big numbers · Multiplication · Division · Factors/Multiples · Order of Operations · Fractions · Mixed Numbers · Decimals · Angles · Area/Perimeter · Volume · Coordinates
This is your chance to show how much you remember!

1. Write $40,000,000 + 300,000 + 5,000 + 70 + 2$ in standard form:

2. Round 13.847 to nearest tenth:

3. $8,040,000 - 3,857,621 =$

4. $648 \times 75 =$

5. $7,344 \div 48 =$

6. $\text{GCF}(36, 54) =$

$\text{LCM}(9, 15) =$

7. $4 + 3 \times (8 - 2)^2 \div 9 =$

8. $\frac{5}{6} + \frac{3}{8} =$

9. $5\frac{1}{4} - 2\frac{5}{6} =$

10. $\frac{3}{4} \times \frac{8}{9} =$

11. $7.4 - 2.863 =$

12. $3.6 \times 0.25 =$

Word Problems

13. A swimming pool is 50 m long, 25 m wide, and 2.4 m deep. What is its volume? If water costs \$0.04 per cubic meter, how much does it cost to fill the pool?

Week 5 Quiz – Geometry & Measurement

W5
QUIZ

Topics: Angles · Shapes · Area/Perimeter · Volume · Coordinates + Cumulative

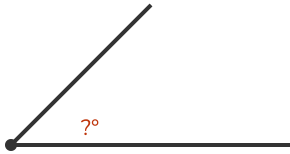
Name: _____

Date: _____

Score: _____ / 20

Section 1 – Angles (5 pts)

1. Measure this angle with your protractor:



Measure:

_____ ° Type: _____

2. Two angles form a straight line. One is 67° . Find the other.

_____ °

3. Triangle angles: 52° , 78° , ?
? =

_____ °

4. Draw a 55° angle using the ray below:



5. **(1 pt)** A quadrilateral has angles 88° , 112° , 95° . Find the fourth angle.

_____ °

Section 2 – Area, Perimeter & Volume (9 pts)

6. Rectangle: $l = 13.5$ cm, $w = 7$ cm
Area:

_____ Perimeter: _____

7. Triangle: base = 10 m, height = 6.4 m
Area:

_____ m^2

8. **(3 pts)** A room is 15 ft long, 12 ft wide, and 9 ft tall. Find: (a) the area of the floor, (b) the volume of the room, (c) how much it costs to carpet the floor at $\$4.50/\text{ft}^2$.

Floor area: